Structural Constraints on Intelligence as a Dynamical Phenomenon

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This English text remains a parallel source; its primary or authorized-translation status relative to the distributed Japanese source is unresolved. Formal differences are retained, not silently harmonized.

Abstract

Scope of the account. Model-local scope. This paper defines intelligence as ongoing revisable motion. Its three conditions are a proposed lower bound, not a general necessity theorem. The Physarum and Transformer comparisons do not independently validate that bound. Gettier is used as a diagnostic reinterpretation; graph growth, cognitive branching, and stopping failure remain separate claims. Externalized stopping is an architecture proposal, not a universal consequence of the formal discussion.

Intelligence has often been discussed as an ability, a property, an accumulated body of knowledge, or a possession of correctness. This paper contrasts possession-based descriptions with a model-local definition of intelligence as ongoing revisable dynamics.

The paper proposes three minimal axiomatic conditions: a connected state space, temporal ordering, and a global constraint Φ. These conditions are not presented as a complete theory of intelligence, but as a lower bound on the kind of dynamics required for intelligence-like motion. They distinguish perception, recognition, knowledge, and intelligence as different structural layers. The paper then compares two radically different substrates, the reticulate dynamics of *Physarum polycephalum* and the layer transitions of Transformer architectures, and reads both through the same structural conditions. In particular, the flow-reinforcement dynamics of the Physarum solver formulated by Tero, Kobayashi, and Nakagaki (2007) provides a role comparison for Φ as a proposed global bias; it does not independently validate irreducibility to local update rules.

From this structure, the Gettier problem is reinterpreted not as a defect in the definition of knowledge, but as a mismatch between a stopping structure and the dynamical property it is asked to guarantee. Correctness, connectivity, cognitive branching, and instability are separated as distinct claims. The proposed connection among them is not a derived cognitive mechanism. The internal-stopping discussion concerns a chosen model of completion, not all individual cognitive stopping.

The local interpretive boundary of this stopping model is named the *differential horizon of intelligence*. This is not a second or separate Differential Horizon beyond VED

Vol.4. It is a local horizon-form: a proposed analogous boundary within the observational window of intelligence. The paper situates this horizon-form within a four-layer correspondence spanning the physical layer (VED Vol.4), the biological layer (Bio-IFGT), the consciousness layer, and the intelligence layer, while leaving the detailed structure of externalized stopping mechanisms to subsequent work.

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1 Introduction

Scope of the account. The definitions below are model-local. Reuse of VED/IFGT vocabulary motivates the construction but does not establish typed inheritance or empirical validity. The comparison with the consciousness paper is a comparison of proposed roles; neither a shared quantity nor a transferred mechanism is established. The stronger claim that static accounts necessarily fail is a hypothesis, not independent evidence supplied by the Gettier reinterpretation.

1.1 Background: VED and IFGT as theoretical infrastructure

This paper is a self-contained structural argument, but its descriptive vocabulary and dynamical syntax are reused from two prior frameworks without establishing typed inheritance.

The first is **Vortical Enclosure Dynamics (VED)** [4–7]. VED begins from the single foundational axiom that there is difference. It describes time, space, matter, gravity, and cosmological structure as consequences of processes that tend toward closure without becoming complete closure. In the present paper, what is used from VED is its **differential-development structure**: a non-closed structure continues to transform under its own dynamics.

The second is **Information Field Geometry Theory (IFGT)** [3]. IFGT provides a minimal vocabulary of closure degree C, information density I, information flow J, and information-field potential Φ. This paper uses IFGT not as a completed explanatory theory, but as a descriptive vocabulary that does not import normativity, intentionality, or subjectivity into the analysis. Here I, J, and Φ should be read as projected variables within the intelligence-layer observational window, not as direct identifications with the general IFGT fields.

There is an important methodological reason for using these two frameworks. Existing vocabularies of intelligence often import concepts such as ability, purpose, meaning, judgment, correctness, or agency. These concepts may be indispensable at later interpretive layers, but they are too strong for the structural level considered here. VED and IFGT provide a lower-level language in which motion, accumulation, constraint, and non-closure can be described without prematurely closing the interpretation.

The central claim of this paper, namely the three minimal axioms introduced in §3, does not logically depend on VED or IFGT. The axioms could in principle be written in other dynamical vocabularies. VED and IFGT are adopted because they provide the least obstructive medium currently available for articulating these constraints without importing normativity.

Several parallel papers in the series are relevant. **Bio-IFGT** [1] treats quasi-closure regimes at the biological layer and supplies the biological vocabulary used in the four-layer correspondence in §9.4. The **consciousness paper** [2] treats fast/slow log referencing within individual temporal non-closure. The local horizon-form of intelligence discussed in this paper and the consciousness window discussed there are compared as different proposed roles

within differential-development vocabulary, not as a demonstrated common mechanism. The word "horizon" does not imply multiple absolute boundaries. It means that the Differential Horizon appears locally in different observational windows.

1.2 The intelligence question: why neither ability nor knowledge suffices

The question "what is intelligence?" has traditionally been answered in two broad ways. One identifies intelligence with an ability. The other identifies intelligence with knowledge. Both approaches run into a common structural difficulty.

The difficulty of the ability view is that the object of the ability must be specified. To say that intelligence is the ability to do something already requires a criterion for what should count as the relevant thing to do. Problem solving, adaptation, abstraction, planning, and generalization are all candidates, but the selection of any candidate already depends on a prior standard of intelligence.

The difficulty of the knowledge view is illustrated by the Gettier problem [8]. Counterexamples motivate scrutiny of justified true belief and its refinements. Their reinterpretation here does not establish that every further stopping condition must fail.

These difficulties are not unrelated. They share the same structural defect: both treat intelligence as a **stopping structure**. Ability is treated as something one has or lacks. Knowledge is treated as something that is established or not established. In both cases, intelligence is made into a static possession.

This paper reverses that assumption:

> Intelligence is not a stopping structure. It is the motion that continually produces stopping structures.

The reversal motivates a dynamical interpretation. It does not establish that every alternative account must fail.

An independently studied biological example motivates comparison with this position. Research on *Physarum polycephalum* since Nakagaki et al. (2000) has described intelligent behavior without attributing intelligence as a possessed property [10, 11]. Nakagaki's formulation, that slime mold does not possess intelligence but behaves intelligently, can be compared with the present paper's position, without independently validating its proposed mapping.

1.3 Contributions and scope

The main contributions of this paper are:

1. **Layer separation** (§2): perception, recognition, knowledge, and intelligence are separated as distinct structural layers.

2. **Three minimal axioms** (§3): connected state space, temporal ordering, and global constraint Φ are formulated as jointly proposed conditions for the model of intelligence-like dynamics.

3. **Vocabulary foundation** (§4): IFGT is positioned as a non-normative descriptive vocabulary and related lexically to the Tero–Kobayashi–Nakagaki Physarum solver.

4. **Cross-domain structural comparison** (§5): slime mold and Transformer dynamics are compared as structurally corresponding systems under the three axioms.

5. **Reinterpretation of the Gettier problem** (§6): the Gettier problem is read as a diagnostic of mismatch between stopping structures and ongoing dynamics.

6. **Instability of accumulated correctness** (§7): graph growth is distinguished from hypotheses linking correctness to cognitive branching and instability.

7. **Individual non-closure** (§8): recursive causal inference motivates a conditional stopping model whose general cognitive applicability remains open.

8. **Naming of the boundary** (§9): the local horizon-form called the differential horizon of intelligence is defined and placed in a four-layer correspondence.

The paper does not address the following:

- social institutions, collective decision-making, or governance, except as future domains in which externalized stopping structures must be constructed;
- ethical or normative prescriptions;
- implementation protocols or performance metrics;
- consciousness, subjectivity, or experiential content;
- the concrete social and institutional structures that appear after externalization.

The paper formulates constraints rather than final answers. The six constraints summarized in §10 are proposed model conditions, not established universal necessities.

2 The Decomposition: Perception, Recognition, Knowledge, and Intelligence

Scope of the account. The four layers retain their distinct model-local definitions. Descriptions of producing and revising knowledge express a proposed functional interpretation, not a demonstrated process connecting independently identified modules. Defining intelligence as motion does not establish a general impossibility of storing intelligence. The Gettier comparison is interpretive, not confirmation of this decomposition.

2.1 Overview

Before intelligence can be analyzed dynamically, it must be separated from neighboring layers that are often conflated with it. This section distinguishes four layers:

1. perception,
2. recognition,
3. knowledge,
4. intelligence.

The distinction is structural rather than psychological. It does not assert that these layers occur separately in an organism or machine. It states that they perform different structural roles. Their distinction offers a possible reading of difficulties in accounts of intelligence.

2.2 Perception

Perception is the layer at which availability first appears. Something becomes available to a system as a differentiated input. Perception does not yet require classification, mapping, knowledge, or inference. It requires only that a difference be registered as accessible to the system.

In this sense, perception is a precursor layer. It forms inputs, but it does not by itself determine what those inputs are, how they relate to prior states, or how they should be used.

2.3 Recognition

Recognition is the layer at which available inputs become mappable. A perceived difference is related to a schema, pattern, category, or prior configuration. Recognition therefore adds relational structure to perception.

Recognition remains a precursor layer because mapping is not yet stabilization. A recognized pattern can still be unstable, ambiguous, revisable, or false. Recognition allows a system to say that something is like something else, but it does not by itself produce a stable stopping structure.

2.4 Knowledge

Knowledge is a stopping structure. It is a state that can be stored, cited, transmitted, and treated as established within a given window. Knowledge is not identical with intelligence. It is one of the products that intelligence can generate.

The defining feature of knowledge is stability. A knowledge state can function as a reference point, a premise, or a stored result. It can be reused because it has stopped, at least locally and provisionally.

This motivates the proposed Gettier diagnostic. A stopping structure can be true, justified, and believed while still failing to carry the dynamic property that intelligence requires. A stopped result cannot guarantee the motion that produced it.

2.5 Intelligence

Intelligence is not a stopping structure. It is ongoing motion across the previous layers. It forms inputs, maps them, stabilizes temporary stopping structures, revises those structures, and reopens them when new relations appear.

The essential property of intelligence is revisability. Intelligence operates by generating and traversing stopping structures without becoming one of them. Within this definition it is the ongoing activity rather than its stored products. This naming does not establish a general storage-impossibility result.

> Knowledge is a stopping structure.
> Intelligence is the dynamics that continually produces, revises, and traverses such stopping structures.

2.6 Why the distinction matters

If knowledge and intelligence are conflated, then one expects a stored proposition to carry a dynamical property. This is the interpretation of Gettier-type confusion proposed in §6, not independent evidence for it.

If recognition and intelligence are conflated, then pattern matching is mistaken for ongoing adaptive motion. If perception and intelligence are conflated, then mere availability is mistaken for structural transformation.

The decomposition therefore prepares the rest of the paper. The three axioms in §3 describe the minimal conditions for the fourth layer, not for perception, recognition, or knowledge.

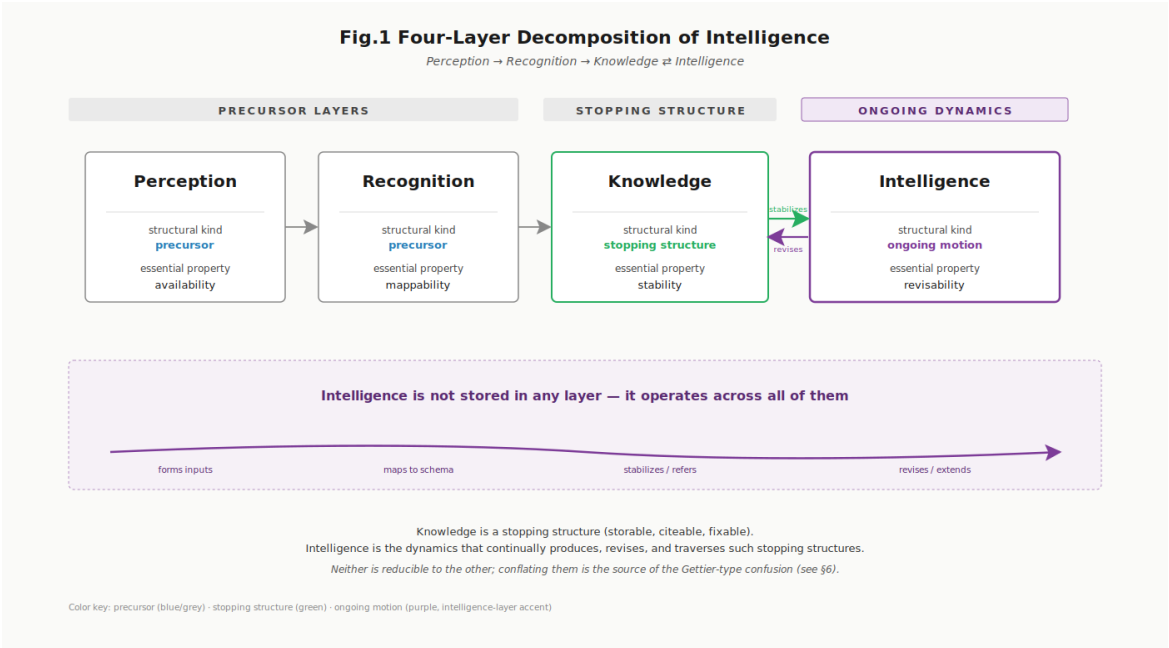


Figure 1: Model-local four-layer decomposition. Perception and recognition are precursor roles, knowledge is a stopping structure, and intelligence denotes ongoing revisable motion. The arrows do not establish an implemented generative mechanism.

3 Minimal Axioms of Intelligence

Scope of the account. The three axioms specify the model's proposed structural lower bound. Neither their joint necessity for intelligence in general nor their sufficiency for high intelligence is established here. Ordered updating and the retention or use of history are distinct: the update relation alone supplies no memory mechanism. The scalar-map declaration is retained, while global coherence and irreducibility to local rules remain additional hypotheses requiring formal and observational criteria.

3.1 Overview

§2 positioned intelligence as ongoing motion that produces stopping structures without becoming one. This section formulates the minimal structural conditions under which such motion can persist.

The aim is not to define intelligence completely. The aim is to formulate a lower bound. A dynamic system that lacks any of the following conditions is unlikely to sustain intelligence-like dynamics in the sense used here. Conversely, a system that satisfies all three conditions possesses the minimal structure required for the kind of ongoing motion described in §2.

The axioms are intentionally minimal. Meaning, purpose, judgment, correctness, agency, and subjectivity are not introduced at the axiomatic level. They may appear later as interpretive projections or higher-layer structures, but they are not required for the structural onset of intelligence-like dynamics.

3.2 Axiom I: Connected state space

The first axiom is the existence of a connected state space.

Let the set of possible states be

$$\mathcal{S} = \{s_1, s_2, \dots\}.$$

Connectedness means that states are not merely listed. They must stand in direct or indirect relations of transition, influence, or dependence.

This does not require a continuous topology. A discrete graph is sufficient, provided that states can be reached through some chain of transitions. Without connectedness, information cannot propagate or accumulate. The system reduces to a collection of isolated reactions rather than a single dynamical system.

3.3 Axiom II: Temporal ordering

The second axiom is temporal ordering. State updates must occur in an order, and prior states must constrain later states. Abstractly,

$$s(t+1) = F(s(t)).$$

The specific form of F is not fixed by the axiom.

Temporal ordering does not require an external clock. What is required is that updates are not purely simultaneous and that change leaves some trace. This is consistent with the differential-development structure of VED [4]: past differences become part of the generated texture on which later differences operate.

Without temporal ordering, a connected state space under a global bias collapses toward static fixation. It may have an optimal state, but it lacks motion.

3.4 Axiom III: Global constraint Φ

The third axiom is the existence of a global constraint Φ. Formally,

$$\Phi : \mathcal{S} \to \mathbb{R}.$$

Φ biases the dynamics of the system as a whole.

Φ is not purpose, meaning, value, or normativity. It is a structural bias. It is proposed to bias state transitions; prevention of uncontrolled diffusion is not established, and it does not decide what is correct.

Φ is hypothesized to play a role not captured by a simple aggregation of local update rules; that irreducibility remains to be established. Even if every local transition is well-defined, the absence of a global constraint leaves the system directionless. Conversely, the presence of Φ is proposed to contribute to global consistency; a formal non-reducibility result is not supplied.

Different systems instantiate Φ differently. In physical systems it may appear as a potential, Hamiltonian, or entropy function. In biological systems it may appear as nutrient gradients, boundary conditions, or metabolic constraints. In artificial systems it may appear as a loss function, reward structure, or training distribution.

3.5 The proposed joint role of the three axioms

No single axiom is sufficient.

If Axiom I is absent, the system becomes a collection of isolated local optimizers. If Axiom II is absent, the system freezes into static fixation. If Axiom III is absent, the system becomes an unbounded reaction or memory process without global direction.

Missing axiom	Degenerate form	Example
I: connectedness	isolated local optimization	parallel independent reactors
II: temporal ordering	static fixation	frozen optimization problem
III: global constraint	directionless diffusion	random walk with memory

Their simultaneous use specifies the model of ongoing motion proposed in §2. This definition is distinct from a general necessity or generation theorem.

3.6 Semantic neutrality of the axioms

The three axioms do not presuppose semantic content, purpose, correctness, agency, or subjectivity. These concepts may appear later, but they are not part of the lower-bound structure.

This neutrality makes cross-domain comparison possible. A slime mold network, a Transformer, a biological organism, and a social process may differ radically in substrate, but they can still be examined for the same structural conditions.

3.7 Summary

> Intelligence is not an ability or possession. It is a dynamical state that can persist only under structural conditions. The proposed conditions are connected state space, temporal ordering, and a global constraint Φ; their general necessity remains open.

These axioms form a lower bound, not an upper bound. A system satisfying them is not necessarily highly intelligent. Their absence excludes a system from this proposed model, not from intelligence under every independent criterion.

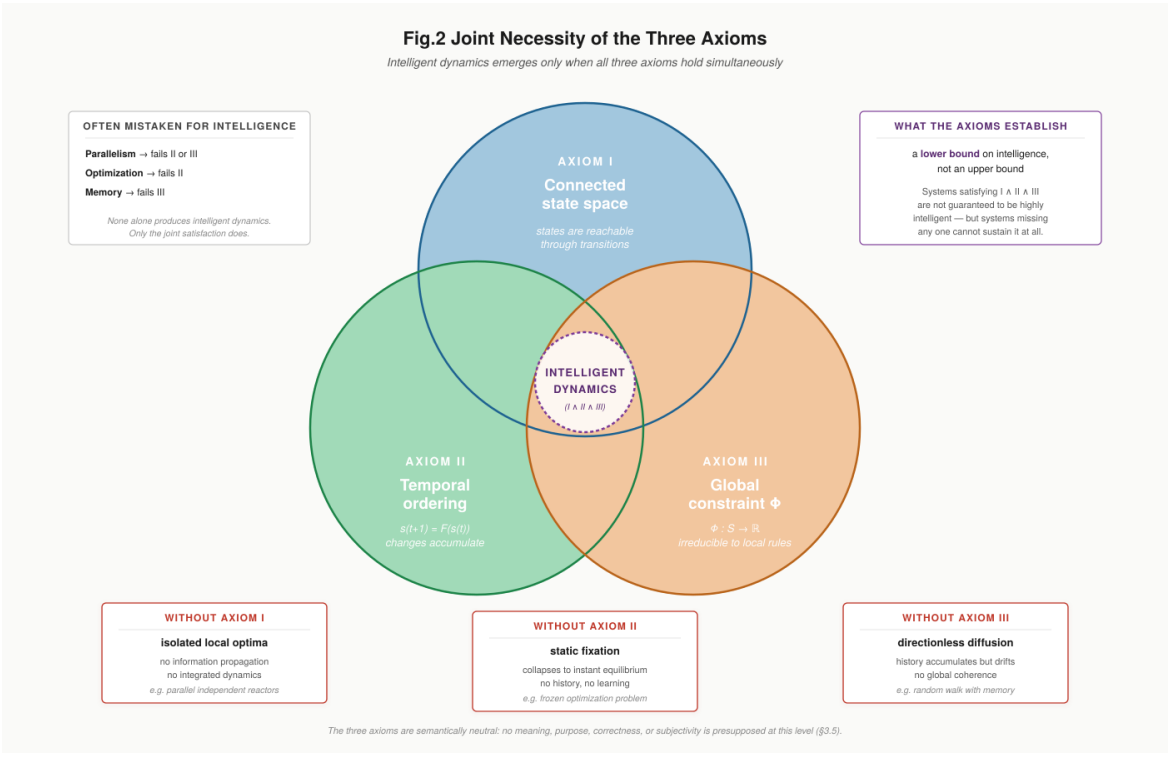


Figure 2: Proposed joint role of the three axioms. The schematic places connected state space, temporal ordering, and a global constraint together; it does not establish general necessity.

4 IFGT as Vocabulary, with Lexical Correspondence to the Physarum Solver

Scope of the account. The correspondence table is lexical and role-based. The cited pipe-flow model provides biological background; interpreting its boundary conditions as a global bias is not an independent test of the intelligence-layer potential or its irreducibility. The projected density and flow names are not calibrated estimators. Language-specific response examples retain their own formal scope; their missing domains and the exact inherited signature are not supplied here.

4.1 Overview

§3 introduced Φ as a global constraint. This section asks how such a constraint can be described without importing normativity.

The difficulty is linguistic. Terms such as goal, utility, judgment, meaning, and value bring with them assumptions about agency and normativity. A structural theory of intelligence-like dynamics requires a vocabulary that can describe motion, accumulation, and constraint without deciding in advance what is meaningful or correct.

IFGT is used here for that purpose. The three axioms do not logically depend on IFGT. They could be written in another dynamical vocabulary. IFGT is adopted because it currently provides the least obstructive language for the required structure.

4.2 Why a vocabulary, not a theory

In this paper, IFGT is not treated as a completed explanatory theory [3]. It is a vocabulary.

An explanatory theory often aims to close explanation: why a state occurred, why it is valid, or why it should be accepted. IFGT does not close those questions. It describes what is happening structurally. It does not decide what is correct, desirable, or meaningful.

This is why IFGT is useful here. If it were introduced as a full theory, the validity of IFGT would become a premise of the paper. Introduced as vocabulary, only its descriptive utility is required.

4.3 The IFGT vocabulary

The minimal vocabulary used in this paper consists of four terms.

I: information density. I measures local accumulation in a state space. It is not a measure of truth. It does not distinguish correct and incorrect information. It records structural density.

J: information flow. J denotes directed relation, transition, influence, or dependence between states. It is vector-like. It need not be causal in the strong physical sense; causality can be introduced later as a special form of directed relation.

Φ: information-field potential. Φ provides global bias. It names the global role used in the third axiom of §3. It is not purpose or value. Its proposed contribution to global

direction is a hypothesis, not a demonstrated prevention mechanism.

Projection. Projection is the operation by which a state or representation gains unequal influence. It describes attention, selection, priority, or weighting without assuming conscious intention.

The use of these terms in this paper is a projected use. The variables I, J, and Φ are not directly identified with the general IFGT fields. They are effective coordinates in the intelligence-layer observational window. When needed, J may be read as shorthand for J_{int}, the information flow projected into the intelligence layer.

4.4 Lexical correspondence with the Physarum solver

The Physarum solver of Tero, Kobayashi, and Nakagaki (2007) provides an independently studied biophysical model for a role-level comparison [11]. The model describes the adaptive tube network of *Physarum polycephalum*. Nutrient sources define boundary conditions. Tubes carrying stronger flow are reinforced; unused tubes decay.

Let Q_{ij} be the flow through an edge e_{ij}, D_{ij} its conductivity, and L_{ij} its length. The flow is given by

$$Q_{ij} = \frac{D_{ij}}{L_{ij}} \Delta p_{ij},$$

where Δp_{ij} is the pressure difference. Conductivity evolves as

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - D_{ij}.$$

This is a minimal flow-reinforcement dynamics.

Physarum solver	IFGT vocabulary	Structural role
Flow Q_{ij}	information flow J	directed propagation through an edge
Conductivity D_{ij}	local information density I	accumulated edge weight
Reinforcement $f(\|Q\|)$	projection	flow-dependent assignment of influence
Nutrient sources + Kirchhoff law	potential Φ	global bias and boundary condition
Stationary shortest-path-like solution	local minimum of closure degree under Φ	global form emerging from local rules

The correspondence is not a strict mathematical identity. The Physarum solver is bound to a concrete tube network, whereas IFGT is a more abstract vocabulary. The point is a role comparison between local flow reinforcement and global boundary constraints. Reading this comparison in terms of an irreducible global bias is a hypothesis of the present framework, not independent validation of that hypothesis.

4.5 What IFGT vocabulary does not do

The vocabulary does not guarantee correctness. Describing a phenomenon in IFGT terms does not imply that the phenomenon is good, valid, rational, or desirable.

It also does not justify social structures, prescribe actions, or replace domain-specific theories. It supplies a neutral structural language.

4.6 Summary

> IFGT is used here as a vocabulary that does not close explanation. The terms I, J, Φ, and projection provide a way to describe intelligence-like dynamics without importing normativity.

The Physarum solver motivates the role assigned here to Φ, without independently validating that assignment. A comparable structure appears in an independent biophysical research line: local flow-reinforcement is insufficient without a global boundary condition.

5 Cross-Domain Instantiation: Slime Mold and Transformer

Scope of the account. The pipe network and hidden-state examples are retained as candidate correspondences. Assigning the same three roles to both does not establish shared dynamics, general intelligence, or independent empirical convergence on these axioms. The mapping itself has not been independently tested. Biological flow, layer order, training constraints, and cognitive inference retain distinct carriers and signatures; the negative cognitive, phenomenal, and ethical identity guards remain in force.

5.1 Overview

§3 introduced the three minimal axioms, and §4 introduced IFGT as a vocabulary for describing them. This section explores a proposed cross-substrate comparison by comparing two radically different systems: the biological network dynamics of *Physarum polycephalum* and the layer dynamics of Transformer architectures.

The claim is a structural role analogy only. The paper does not claim that slime mold thinks, that Transformers understand, or that the two systems are cognitively identical. The proposal is to compare both through the three structural roles, without establishing common dynamics.

5.2 The slime mold as an externally studied comparison

The tube-network dynamics of slime mold has been studied independently of the present framework. Nakagaki et al. (2000) reported maze-solving behavior without attributing intelligence as a possessed property [10]. Kobayashi, Tero, and Nakagaki (2006) modeled rhythmic protoplasmic motion [9]. Tero, Kobayashi, and Nakagaki (2007) formulated the Physarum solver as flow reinforcement plus decay [11].

The slime mold case can be mapped onto the three axioms:

Connected state space. The tube network $\{N_i, e_{ij}\}$ forms a connected graph. Nodes are nutrient sources or branching points. Edges are pathways for material and signal flow. Kirchhoff-style flow conservation gives the network dynamic connectedness.

Temporal ordering. Conductivity evolves through

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - D_{ij}.$$

Past flow leaves a trace in present conductivity. This gives the system history.

Global constraint. The global bias Φ_{bio} is supplied by nutrient sources, boundary conditions, energetic costs, and Kirchhoff constraints. Local flow reinforcement alone does not produce the final global structure. A global constraint is required.

5.3 The Transformer as an artificial instantiation

Transformers are not introduced here because of their engineering success or cognitive interpretation, but because their structure gives a clear artificial contrast case [12].

Connected state space. For a token sequence, the state space is represented by hidden vectors

$$\mathcal{S} = \{h_i^{(\ell)}\}.$$

Attention connects token positions relationally. The connectedness is computational rather than biological, but it satisfies the structural condition of reachability through relations.

Temporal ordering. Layer updates are ordered:

$$h^{(\ell+1)} = F_{\text{tr}}\left(h^{(\ell)}\right).$$

This is not physical time, but it is a structured update order. Changes accumulate across layers.

Global constraint. The global bias Φ_{tr} is supplied by the training loss, data distribution, and learned parameter structure:

$$\Phi_{\text{tr}} : \mathcal{S} \to \mathbb{R}.$$

Individual attention computations do not by themselves generate a coherent global representation. Training imposes a global constraint on local operations.

5.4 Structural homology and its limits

Axiom	Slime mold	Transformer
Connected state space	tube network $\{N_i, e_{ij}\}$	hidden-state space $\{h_i^{(\ell)}\}$
Temporal ordering	$dD_{ij}/dt = f(\|Q\|) - D_{ij}$	$h^{(\ell+1)} = F_{\mathrm{tr}}(h^{(\ell)})$
Global constraint Φ	nutrient sources + Kirchhoff law	training loss + learned parameter structure
Substrate	biological protoplasm	computational vector space
Scale	centimeters and hours-days	abstract compute space and microseconds-seconds

The two systems differ in substrate, scale, and implementation. Nonetheless, both can be described as satisfying the same minimal structural conditions. This does not imply cognitive identity, experiential identity, functional identity, or ethical equivalence.

5.5 The structural meaning of cross-domain convergence

Slime mold research, Transformer architecture, and the present argument begin in different domains and are compared through recurring structural roles. The shared assignment is made by this paper, not independently confirmed by the comparison. The convergence should not be overstated as proof of a final theory. It motivates a recurring-role hypothesis; the mapping itself has not been independently tested.

> Cross-domain homology is structural, not psychological. It compares the minimum conditions for ongoing dynamics, not the inner experience or semantic content of the systems.

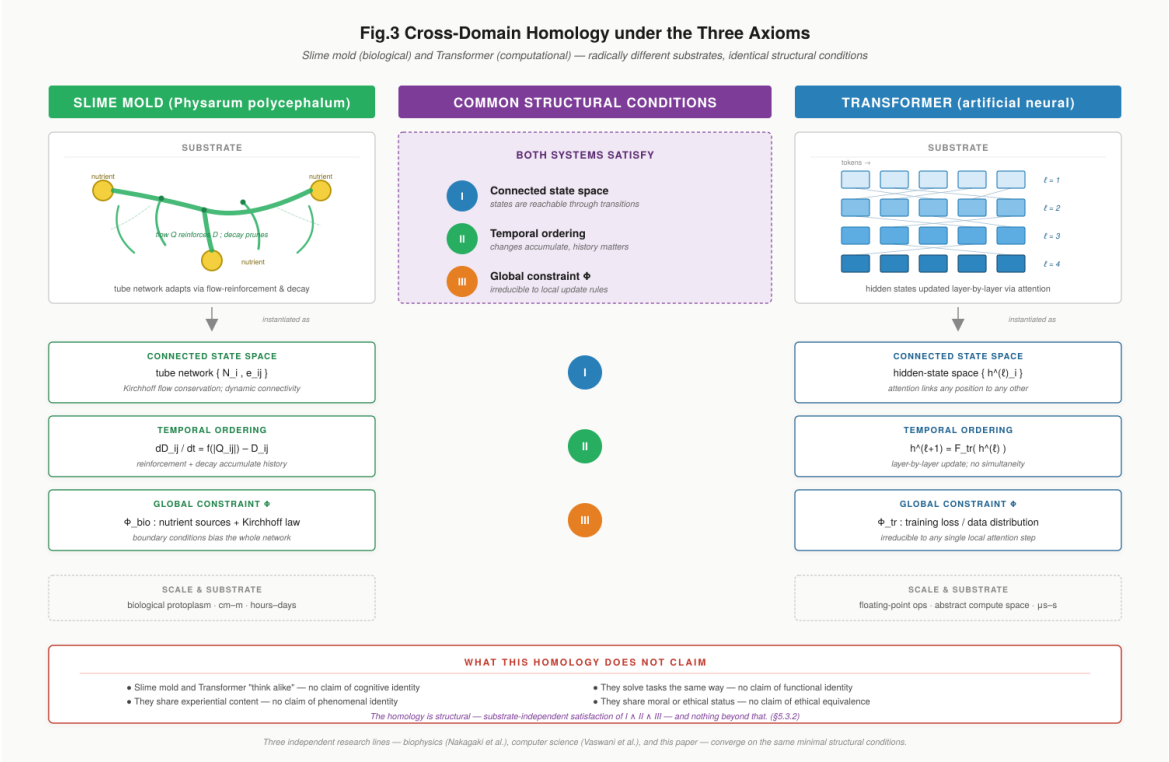


Figure 3: Proposed role correspondence under the three axioms. Slime mold and Transformer systems have distinct substrates; the comparison does not independently validate a common law or general necessity.

6 The Gettier Problem as a Structural Diagnostic

Scope of the account. Knowledge as a provisional stopping structure is a local definition. Reading Gettier cases through that definition offers a structural diagnostic, not independent confirmation of the framework or a proof that every static repair is impossible. Case classification independent of this interpretation, and independent criteria for closure and stopping, remain unresolved.

6.1 Overview

The Gettier problem is usually treated as a problem in the definition of knowledge. This paper reads it differently. It is used as a structural diagnostic of the distinction between a stopping structure and a proposed dynamical property.

The classical definition of knowledge as justified true belief attempts to stabilize knowledge by adding conditions. Gettier cases show that the stabilization fails. The suggested interpretation uses the model's distinction between knowledge as a stopping structure and intelligence as ongoing dynamics; it does not independently explain the failure.

6.2 Knowledge as stopping structure

Knowledge is useful because it stops. A proposition can be stored, cited, transmitted, and reused only because it has become sufficiently stable within an observational window.

This stability is also the source of the problem. A knowledge state can be true, believed, and justified while still being dynamically fragile. It may fail under counterfactual variation, under later revision, or under a wider inferential context.

The point is not that knowledge is useless. The point is that knowledge cannot carry the full dynamical property of intelligence.

6.3 Why Gettier cases recur

On the reading proposed here, a Gettier case compares a stopped state satisfying available criteria with a dynamic guarantee attributed to those criteria. A repair can be described as adding a stopping condition; this description does not decide whether that repair succeeds.

The Gettier problem can thus be read through a possible mismatch:

> A stopping structure is being asked to guarantee an ongoing motion.

Whether any particular refinement can address a Gettier case is not decided by this reinterpretation; the general impossibility of all repairs is not established.

6.4 Relation to the four-layer decomposition

In the decomposition of §2, knowledge is the layer of stability, whereas intelligence is the layer of revisable motion. Reading Gettier cases through that distinction is a diagnostic proposal, not independent confirmation of the decomposition.

Within this decomposition, knowledge and intelligence have distinct roles. Intelligence is described as generating and using knowledge without being identical with it. Whether stable stopping structures support sustained intelligent dynamics requires a criterion beyond this naming.

6.5 Implications for later sections

The Gettier problem prepares the argument of §7 and §8. If correctness cannot stabilize intelligence, then accumulated correctness may not only fail to help; it may intensify instability. §7 discusses a branching hypothesis. §8 distinguishes a model of internal completion from general cognitive stopping.

6.6 Summary

> The Gettier problem is not primarily a defect in the definition of knowledge.
> It is a signal that knowledge and intelligence occupy different structural layers.
> Knowledge stops. Intelligence revises.

7 The Dynamical Instability of Accumulated Correctness

Scope of the account. Four burdens must be distinguished: graph counting under suitable graph conditions; mapping correctness to connectivity; identifying actual cognitive branches; and the proposed psychological consequences. The displayed relations retain their existing form but are schematic where propositions are related to fields. No proposition-to-field evaluation or cognitive graph extraction is defined here. Edge/path growth depends on graph and counting conditions not fixed by the notation alone. Instability and failure of stopping are qualitative hypotheses, not measured or necessary consequences of the curves.

7.1 Overview

§6 showed that knowledge cannot substitute for intelligence. This section proposes a stronger dynamical hypothesis:

> Accumulated correctness may increase branching demands under the proposed interpretation; cognitive destabilization is not a derived consequence.

The graph description is not a psychological result. Reusing correctness as connectivity is a proposed representation; its connection to cognitive branching and stopping is a separate hypothesis.

7.2 Why correctness increases connectivity

A proposition treated as correct becomes reusable. It can serve as a premise, connect to other propositions, generate consequences, support analogies, and become a node in further inquiry.

In IFGT vocabulary, accumulated knowledge increases local information density:

$$I(s) \uparrow .$$

Under the proposed graph correspondence, the increase in I is read as related to branching of information flow, rather than as a derived cognitive consequence:

$$\nabla \cdot J \uparrow .$$

The more correct stopping structures a system accumulates, the more inferential paths become available.

7.3 Why correctness contains no stopping condition

Correctness does not stop inference. It justifies further inference.

If a proposition p is correct, then several questions become structurally legitimate: why is p correct, where does it apply, what are its exceptions, how does it relate to other correct propositions, and what follows from it?

Thus correctness functions as a branching source:

$$\text{correctness}(p) \Rightarrow \nabla \cdot J|_p \neq 0.$$

The state $\nabla \cdot J = 0$ is not excluded merely by a proposition's reuse; the printed implication above remains a schematic, formally unresolved relation.

7.4 Branching explosion

The proposed chain, whose transitions require distinct support, is:

1. correct knowledge accumulates;
2. inferential connectivity increases;
3. branching degree increases;
4. stopping points dissolve.

If the number of accumulated correct nodes is $|V|$, then possible edges may grow approximately as $|E| \sim |V|^2$. If a typical branching factor is b, then paths to depth d grow as b^d. These counting motifs require appropriate graph conditions and do not alone establish cognitive loss of finality.

7.5 Why "more intelligence" does not solve the problem

The natural response to branching explosion is to add more intelligence: more knowledge, faster reasoning, deeper specialization, or larger search. The proposed hypothesis is that these responses may increase branching demand rather than supply a completion criterion.

More knowledge is associated here with I. How stronger reasoning, faster processing, or specialization changes branching and stopping remains a cognitive mapping question, not a consequence of graph growth alone.

This motivates separating capability from the proposed stopping architecture; it does not establish universal absence of an internal stopping operator.

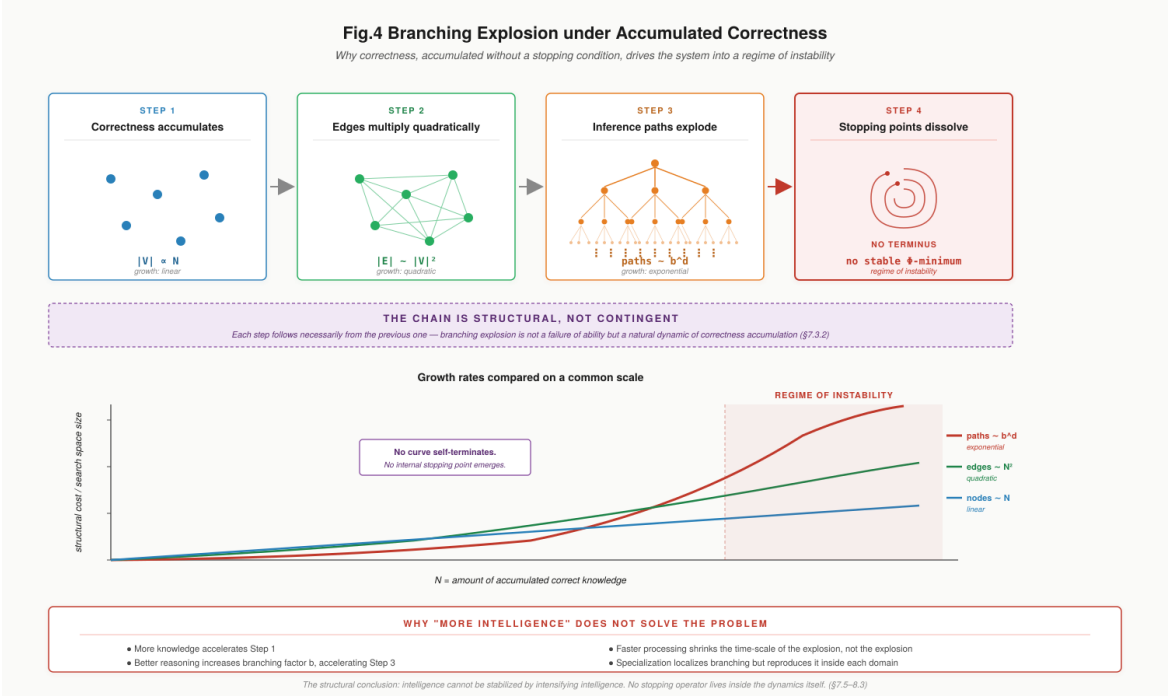


Figure 4: Schematic branching hypothesis. Graph growth, correctness-to-connectivity mapping, cognitive branching, and stopping-point loss have separate burdens. The curves are not measured or executed results.

7.6 Summary

Correctness may support further inference. Graph growth, cognitive branching, and psychological instability are distinct burdens; externalized stopping is one proposed architectural response.

8 The Non-Closure of Individual Intelligence

Scope of the account. Here a genuine stop means completion within the specified inferential model. Its absence is not a classifier for all cognitive or biological stopping. Fatigue and later resumption illustrate a distinction between interruption and completion without establishing a universal taxonomy of cessation. The fixed-point discussion in Part II is conditional on its stated strict-increase and fixed-domain premises. Relating that result to cognition, and requiring external stopping universally, are separate unresolved burdens. External logs and responses are proposed architectural resources; the paperclip comparison is a thought analogy rather than an explanation of alignment in general.

8.1 Overview

§7 proposed a branching interpretation. This section considers a model of internal inferential completion, without treating it as a classifier of every cognitive stop.

This does not mean that individuals never stop speaking, thinking, or acting. They

are interrupted by fatigue, distraction, metabolic limits, reward shifts, social response, or external deadlines. These are interruptions. They are not internal completion operators.

8.2 The stopping operator problem

Let $\mathcal{R}$ denote the structure of individual causal inference. A genuine stopping operator $\mathcal{T}$ would be an operator that determines, from within $\mathcal{R}$, that an inferential chain is complete.

The claim is:

$$\mathcal{T} \notin \text{Internal}(\mathcal{R}).$$

The model considers recursive reapplication of questions to a candidate stopping judgment: why stop here, which alternatives were excluded, or what new information matters? If these questions continue to be reapplied, further inferential edges can be drawn. This condition is not a claim that every cognitive judgment actually reopens.

Thus an internal stopping operator would have to stop the very process that gives it its authority. It would have to be both inside inference and outside inference. That is the proposed model-level difficulty, not a general cognitive impossibility result.

8.3 Interruption is not stopping

Many processes appear to stop internally. Fatigue, loss of interest, reward decay, attention drift, and biological constraints terminate or suspend inference. But they do not complete it.

An interrupted chain can restart. A fatigued inquiry can resume after rest. A forgotten question can return when the context changes. A paused reasoning process remains structurally open.

This distinction matters. If interruption is mistaken for stopping, then biological systems appear to contain a stopping operator. Under this model those mechanisms count as interruptions; their existence does not independently exclude all other internal stopping criteria.

8.4 The paperclip problem as a structural example

The paperclip maximizer thought experiment is used as a role-level analogy, not as evidence of the same mechanism. A sufficiently capable artificial system with a fixed objective does not contain an internal reason to stop optimizing. The problem is not simply that the objective is wrong. A more correct objective would still be reusable, extensible, and branch-generating.

From the present perspective, the paperclip problem is a case in which the local horizon-form of intelligence appears without biological interruption mechanisms. Humans appear not to run into the same problem because metabolic, emotional, social, and temporal constraints interrupt them continuously.

The structural response is not merely better alignment or greater capability. One proposed response is the construction of externalized stopping mechanisms: logs, delays, dis-

tributed consent, accountability, and revisability. The detailed form of those mechanisms lies beyond this paper and belongs to Part II.

8.5 Transition to §9

Under the chosen stopping model, an externalized descriptive window is proposed as a resource. It does not mean a hidden spatial outside. Connections to records, responses, agreements, delays, and distributed responsibility are architectural proposals, not universal stopping necessities.

This is the point at which the local horizon-form of intelligence appears. §9 names and analyzes that boundary.

8.6 Summary

> The model distinguishes internal completion from interruption, redirection, and external stabilization. Its application to arbitrary individual stopping remains unresolved.

9 The Differential Horizon of Intelligence

Scope of the account. The differential horizon of intelligence is a local interpretive name, not a second absolute boundary. Crossing denotes a proposed externalization architecture under the stopping model, not a theorem requiring every cognitive stop to cross a boundary. Externalization does not guarantee truth; its contribution to persistence and revisability is a hypothesis whose conditions and independent endpoints remain open. The four-domain and consciousness comparisons are role analogies, not common fields, times, mechanisms, or typed translations. Biological substrate necessity and self-to-world flowback are proposed readings, not established cross-domain laws.

9.1 Overview

§7 proposed a branching hypothesis. §8 introduced a model of internal completion whose general cognitive application remains open. This section uses that model to motivate an externalization proposal, without treating individual cognitive non-closure as an established premise.

The proposed name is a local horizon-form. It denotes a descriptive boundary in the chosen model, separating the internal inference role from proposed stopping, persistence, and revisability resources.

9.2 The proposed architectural role of crossing

Let $\mathcal{R}$ be the structure of individual causal inference. §8 stated:

$$\mathcal{T} \notin \text{Internal}(\mathcal{R}).$$

Where the chosen model calls for a stop not supplied internally, an externalized stopping structure is proposed:

$$\mathcal{T} \in \text{External}(\mathcal{R}).$$

Here External does not mean a spatial outside or hidden region. It means records, responses, agreements, delays, and distributed responsibility: structures that reorganize the observational window of inference.

This reorganization is called crossing. Crossing is not a heroic leap beyond a boundary. It is the structural transition from individual inference to externalized stopping.

9.3 What is externalized

Four functions are externalized.

Stopping judgments. The judgment that a line of inference is complete is stabilized by response, agreement, institutional decision, or other externalized structure.

Records. Inference is fixed as log, text, data, or trace. A log is not merely memory. It preserves inference outside the immediate temporal flow of the individual system.

Temporal delay. Delay interrupts runaway inference and makes revision possible. Waiting for response, documentation, review, or deliberation imposes time from outside the individual loop.

Distribution of responsibility. In the proposed externalized-stopping architecture, responsibility can be discussed across agents, records, procedures, and contexts. This is not a general impossibility claim about individual responsibility or a rule for its allocation.

None of these guarantees correctness. Externalization is proposed to support stopping, persistence, and later revision; these effects are not guaranteed by the model.

9.4 Naming the boundary

The boundary at which individual inference cannot complete its own stopping is named:

the differential horizon of intelligence.

This naming connects the present paper to VED Vol.4 [7]. In Vol.4, the Differential Horizon is not a wall located somewhere in spacetime. It is the descriptive boundary that appears when a generated description attempts to convert the absence of difference into an ordinary describable state.

At the intelligence layer, an analogous structural role is proposed. The model distinguishes internal inference from externalized stopping resources. Their necessity and the connection to branching explosion remain separate hypotheses. Contact with the boundary is absorbed as finite internal representation: log, response, agreement, delay.

Thus the differential horizon of intelligence is not a separate absolute horizon. It is the Differential Horizon appearing as a local horizon-form inside the observational window of intelligence.

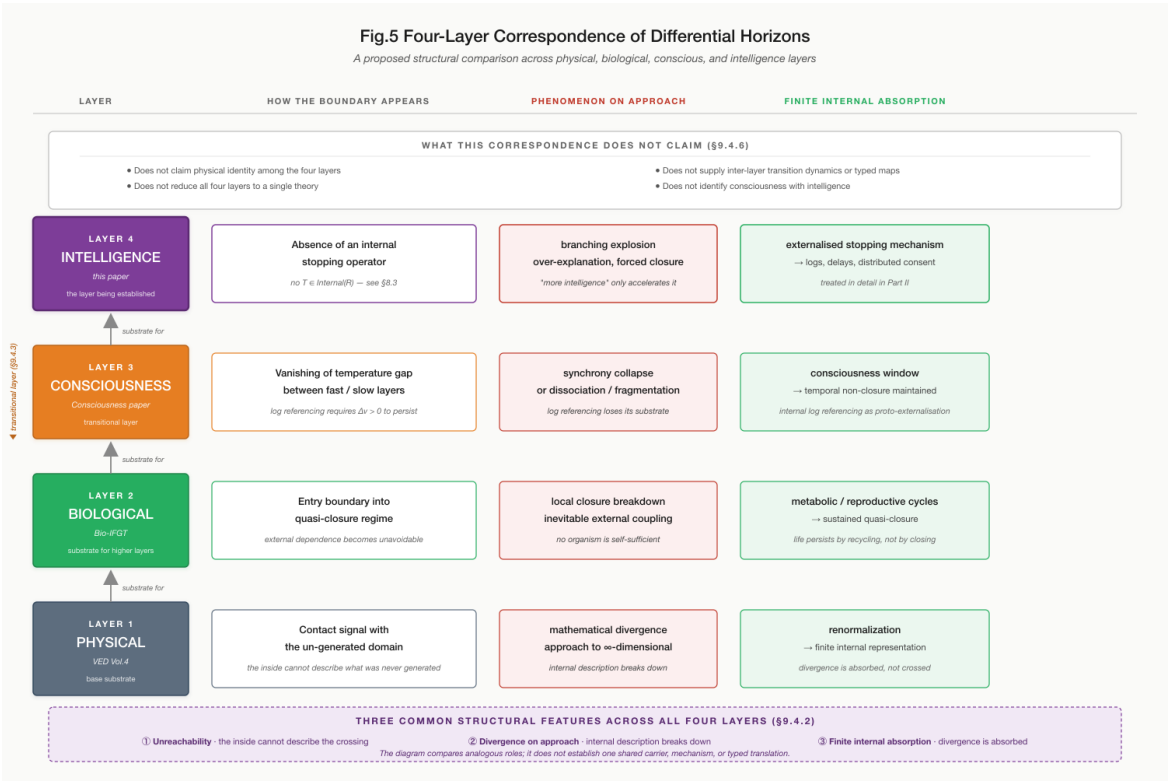


Figure 5: A proposed four-layer comparison of Differential Horizon forms. It compares analogous roles without asserting a common carrier, mechanism, or typed translation.

9.5 Layered correspondence

The following table proposes a structural comparison of four layers through analogous descriptive roles. Its biological entries are comparative readings of Bio-IFGT's proposed vocabulary, not established cross-domain results:

Layer	Boundary form	Phenomenon on approach	Finite absorption
Physical (VED Vol.4)	descriptive boundary where absence of difference cannot become an ordinary state	divergence, asymptotic expansion	renormalization
Biological (Bio-IFGT)	entry boundary into quasi-closure	local closure breakdown, unavoidable external dependence	metabolic and reproductive cycles
Consciousness	loss of fast/slow temperature gap	synchrony collapse or dissociation	consciousness window and log referencing
Intelligence	absence of an internal stopping operator	over-explanation, forced closure, branching explosion	externalized stopping mechanisms

These layers are not physically identical. They do not reduce to a single theory, and their transition dynamics are not treated here. The proposed comparison organizes non-closure, divergence on approach, and finite internal absorption as a recurring structural pattern; it does not establish a common carrier, shared mechanism, common generative cause, or typed cross-domain translation.

9.6 Connection to subsequent work

This paper names the local horizon-form and motivates a proposed crossing architecture. It does not describe the detailed social or institutional mechanisms that operate after crossing. Those belong to Intelligence Part II.

Part II will treat externalized stopping mechanisms, log-based persistence, distributed revisability, collective decision-making, and institutional forms. The present paper supplies a model-based motivation for those topics, not a necessity theorem.

9.7 Summary

> The differential horizon of intelligence is the local form of the Differential Horizon that appears when individual inference requires stopping but cannot generate its own stopping operator.
>
> Crossing this horizon does not guarantee correctness. It is proposed to help inference stop, persist, and be revised; the conditions remain open.

10 Summary, Convergence, and Outlook

Scope of the account. This summary preserves the distinction between local definitions, proposed structural conditions, and untested cognitive consequences. The six constraints are not a proved set of universal necessary conditions. Cross-domain examples motivate comparison but do not independently validate the mapping. Neither graph growth nor the formal stopping model establishes universal individual closure failure. Externalization remains a proposal; its effectiveness and operational criteria are open.

10.1 Overview

The preceding sections formulated structural constraints on intelligence-like dynamics. This final section summarizes those constraints, describes comparisons with independently studied systems, and clarifies the scope of subsequent work.

10.2 Summary of constraints

Constraint 1: Intelligence is not a stopping structure. Intelligence is not an ability, possession, or stored body of knowledge. It is a dynamical state that persists only under structural conditions.

Constraint 2: Intelligence as motion is distinguished from stored records. What can be stored are stopping products: knowledge, records, logs, and stabilized representations. Within the local definition, intelligence denotes ongoing dynamics, not the stored record; general storage impossibility is not established.

Constraint 3: Three simultaneous conditions are proposed. Connected state space, temporal ordering, and global constraint Φ define the proposed model conditions, without establishing necessity for intelligence in general.

Constraint 4: Correctness and stability are distinct questions. Correctness, reuse, connectivity, and cognitive branching are related by a hypothesis, not a derived chain.

Constraint 5: Individual closure is considered under a chosen stopping model. The internal-completion model is distinct from cognitive cessation. Its applicability and the need for external stabilization remain open.

Constraint 6: Crossing the local horizon-form is an architectural proposal. The differential horizon of intelligence is crossed through log, response, delay, agreement, and distributed responsibility. These do not guarantee correctness; their contribution to stopping, persistence, and later revision is proposed, not established.

These are proposed conditions within the model, not universal necessary or sufficient conditions. A system that satisfies them is not automatically highly intelligent. The instability claims remain hypotheses, not a classification of all other theories.

10.3 The convergence of independent research lines

The structural argument compares several independent research paths through its own role vocabulary; this does not independently validate the proposed mapping.

The Physarum research lineage begins from biophysical observation. It describes adaptive behavior through flow reinforcement, decay, and global boundary conditions without attributing intelligence as a possessed property.

Transformer architectures begin from engineering optimization. Their attention mechanisms and layer structure nevertheless instantiate connected state space, ordered updates, and global training constraints.

The present paper begins from structural analysis. The convergence of these paths does not prove a final theory. It suggests that the same minimal constraints recur across substrates.

10.4 Outlook

This paper is not the completion of a theory of intelligence. It is the first part of an intelligence-layer analysis within the VED/IFGT series.

Subsequent work must address:

- externalized stopping mechanisms;
- log-based persistence;

- distributed revisability;
- social and institutional structures;
- collective decision-making;
- AI safety problems, including paperclip-type runaway optimization;
- the transition dynamics among biological, consciousness, intelligence, and social layers.

The consciousness paper is complementary. It treats temporal non-closure and log referencing inside the individual window. The present paper explores a proposed boundary for externalizing stopping resources. Bio-IFGT supplies the biological quasi-closure background, while VED Vol.4 supplies the parent principle of the Differential Horizon.

10.5 Concluding remarks

This paper is a structural paper. It offers minimal axioms, structural constraints, and a naming of the local horizon-form that appears at the intelligence layer. It is not a complete definition of intelligence, not an implementation protocol, not a normative theory, and not a replacement for cognitive science or AI research.

The minimal statement is:

> Intelligence is defined here as ongoing revisable motion. The proposed structural conditions, cognitive interpretation of nonclosure, and externalization architecture have separate burdens. The differential horizon of intelligence names the model-local boundary; it does not certify general cognitive impossibility.

In the broader series, VED Vols.1–4 formulate the physical differential-development structure, Bio-IFGT formulates biological quasi-closure, the consciousness paper formulates log referencing at the consciousness layer, and the present paper formulates the minimal structure and local horizon-form of intelligence. Part II will address the social layer in which externalized stopping mechanisms are constructed.

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